

(Please write your Enrolment No. immediately)

Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

First Semester, November, 2023

Paper Code: BS111

Subject: Applied Mathematics

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Find $\frac{\partial^3 u}{\partial x \partial y \partial z}$ if $u = e^{x^2+y^2+z^2}$.	2.5	CO1
(b)	Calculate $\frac{\partial(u,v)}{\partial(r,\theta)}$, if $u = 2axy$, $v = a(x^2 - y^2)$ where, $x = r \cos \theta$, $y = r \sin \theta$.	2.5	CO1
(c)	Solve $(3x^2 + \frac{y}{x})dx + (x^3 + \ln x)dy = 0$.	2.5	CO2
(d)	Show that family of curves $x^2 + 4y^2 = c_1$ and $y = c_2 x^4$ are mutually orthogonal to each other.	2.5	CO2
2(a)	If $u = \ln(x^3 + y^3 - x^2 y - xy^2)$ then show that $u_{xx} + 2u_{xy} + u_{yy} = -\frac{4}{(x+y)^2}$.	5	CO1
(b)	Find the maximum and the minimum distances from the origin to the curve $3x^2 + 4xy + 6y^2 = 140$.	5	CO1
3(a)	Prove that $\frac{d}{dx}\{x^{-n}J_n(x)\} = -x^{-n}J_{n+1}(x)$.	5	CO2
(b)	Show that $\frac{d}{d\alpha} \int_0^{\alpha^2} \tan^{-1} \frac{x}{2} dx = 2\alpha \tan^{-1} \alpha - \frac{1}{2} \log(\alpha^2 + 1)$.	5	CO1
4(a)	Solve $y(y^2 - 2x^2)dx + x(2y^2 - x^2)dy = 0$.	5	CO2
(b)	Solve $(D^2 + 2D + 1)y = e^{-x} \ln x$.	5	CO2

Mid-Term Examination – February 2022

Programme: B.Tech (All branches)

Semester: First Semester (Dec. 2021 - March 2022)

Paper Code: BS111

Paper Name: Applied Mathematics-1

Time: 1.5 hours

Maximum Marks: 30

Note:

- Question No. 1 is compulsory.
- Attempt any two questions from the remaining questions.
- Some questions have internal choice also.
- All questions carry equal marks.
- Only scientific calculator is allowed.

Question. 1 :

(5 × 2 = 10 Marks)

(i) If $x = r \cos \theta$ and $y = r \sin \theta$, then examine the Jaccobian $J = \frac{\partial(x,y)}{\partial(r,\theta)}$ is.....(ii) If $R = f_{xx}(a,b)$, $S = f_{xy}(a,b)$, $T = f_{yy}(a,b)$, then $f(x,y)$ will have a maximum value at (a,b) if

- (a) $f_x = 0$ $f_y = 0$, $RT - S^2 < 0$ and $R < 0$ at (a,b)
 (b) $f_x = 0$ $f_y = 0$, $RT - S^2 = 0$ and $R > 0$ at (a,b)
 (c) $f_x = 0$ $f_y = 0$, $RT - S^2 > 0$ and $R > 0$ at (a,b)
 (d) $f_x = 0$ $f_y = 0$, $RT - S^2 > 0$ and $R < 0$ at (a,b)

Recognize the right answer.

(iii) The complementary function of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \sin x$ is:

- (a) $C_1 e^x + C_2 e^{-x}$ (b) $(C_1 + x C_2) e^{-x}$
 (c) $(x C_1 + C_2) e^x$ (d) None of these.

Examine the right answer with proper steps.

(iv) Evaluate the particular integral of $(D^2 - 4)y = \sin 3x$ is....., where $\frac{d}{dx} = D$.(v) Examine the solution of the ODE: $\frac{dy}{dx} + y = 0$, for given $y(0) = 5$ is.....

Question. 2 :

(2 × 5 = 10 Marks)

(i) Prove that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.

OR

Examine the maximum and minimum value of $f(x,y) = \sin(x)\sin(y)\sin(x+y)$ (ii) Evaluate $\int_0^1 \frac{x^a - 1}{\log x} dx$, $a \geq 0$.

OR

If $u = x\sqrt{1-y^2} + y\sqrt{1-x^2}$, $v = \sin^{-1}(x) + \sin^{-1}(y)$, show that u and v are functionally related and create a relationship in between.

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Question. 3 :

(2 × 5 = 10 Marks)

- (i) Examine the solution of ODE: $2y dx + x(2 \log x - y) dy = 0$.

OR

Examine the solution of the differential equation: $(D^2 + 4)y = \tan 2x$.

- (ii) Analyze the solution of the Euler-Cauchy ODE: $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = x + \log x$.

OR

Find the solution of the differential equation: $(1 + y^2)dx + (x - e^{-\tan^{-1} y})dy = 0$.

Question. 4 :

(2 × 5 = 10 Marks)

- (i) If $u = f(r)$ and $x = r \cos(\theta)$, $y = r \sin(\theta)$, prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r).$$

OR

Prove the following recurrence formula for Legendre's Polynomial:

$$P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

- (ii) (Fluid Flow) If the stream lines (path of fluid particles) of a flow around a corner are $xy = c$, c is a constant. Construct the orthogonal trajectories (known as equipotential lines).

OR

Prove the following formula for the Bessel function:

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] DECEMBER 2024

Paper Code: BS-111

Subject: Applied Mathematics-I

Time: 3 Hours

Maximum Marks: 60

Note: Attempt any five questions including Q.No1 which is compulsory.
Internal Choice is indicated.

Q1. Attempt any Four of the following questions:

(4x5=20)

a) For the transformation $x = a(u+v)$, $y = b(u-v)$ and

$$u = r^2 \cos 2\theta, v = r^2 \sin 2\theta, \text{ Find } \frac{\partial(x, y)}{\partial(r, \theta)}.$$

b) Test for consistency the following equation and solve them if consistent ✓

$$3x + 7y + 8z = -13, 2x + 9z = 5, -4x + y - 26z = 2$$

c) If vector $\vec{v} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal vector, then ✓
find the value of a.d) Find the directional derivative of $f(x, y, z) = xy^3 + yz^2$ at the point ✓

$$(2, -1, 1) \text{ in the direction of vector } \hat{i} + 2\hat{j} + 3\hat{k}$$

e) If $z = f(x+ay) + \phi(x-ay)$, then show that $z_{yy} = a^2 z_{xx}$

$$f) \text{ Solve } \frac{dy}{dx} = \frac{x(x^2 + y^2 - 1)}{y(x^2 + y^2 + 1)}$$

g) Define Gamma Function and Prove that $\int_0^{\infty} e^{-x^2} dx = \frac{1}{2} \sqrt{\pi}$ h) Verify the vectors $(1, 9, 9, 8), (2, 0, 0, 3), (2, 0, 0, 8)$ are linearly Independent ✓
or not?

Q2

a) If $\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u} = 1$, show that $u_x^2 + u_y^2 + u_z^2 = 2(xu_x + yu_y + zu_z)$. ✓ (5)b) Show that the function $f(x, y) = x^3 \exp(-x^2 - y^2)$ has a maximum at the (5)
point $(\sqrt{3/2}, 0)$. A minimum at $(-\sqrt{3/2}, 0)$ and a stationary point at the
origin whose nature cannot be determined.

OR

Q3 a) Evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$ by applying differentiation under integral sign (5)b) Find the stationary points of $f(x, y, z) = x^3 + y^3 + z^3$ subject to constraint (5)
 $g(x, y, z) = x^2 + y^2 + z^2 = 1$ Q4 a) Solve $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$ (5)

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b) Solve $(x^2 y - 2xy^2) dx = (x^3 - 3x^2 y) dy$ (5)

OR

Q5 a) Solve $(D^2 + 2D + 1)y = 2x + x^2$ (5)

b) Apply the method of variation of parameters, to solve $\frac{d^2 y}{dx^2} + y = \operatorname{Cosec} x$ (5)

Q6 a) Solve the following system by Gauss Elimination method (5)

$$2x_1 - x_2 + 3x_3 = 9$$

$$x_1 + x_2 + x_3 = 6$$

$$2x_1 - x_2 + x_3 = 2$$

b) If $\begin{vmatrix} a & a^2 & a^3 - 1 \\ b & b^2 & b^3 - 1 \\ c & c^2 & c^3 - 1 \end{vmatrix} = 0$ in which a, b, c are different, show that $abc = 1$. (5)

OR

Q7 a) State Cayley-Hamilton Theorem and verify Cayley-Hamilton Theorem for (5)

the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

b) Find the diagonalization for the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ (5)

Q8 a) Using Green's theorem find the line integral $\oint_C (x^2 - 2xy) dx + (x^2 y + 3) dy$ (5)

where C is the boundary of the region defined by $y^2 = 8x$ and $x = 2$

b) Find the directional derivative of $f(x, y, z) = xy^3 + yz^2$ at the point (2, 1, 1) in the direction of vector $\hat{i} + 2\hat{j} + 3\hat{k}$ (5)

OR

Q9. a) Find the divergence and curl of $F = 2x^2 z \hat{i} - xy^2 z \hat{j} + 3yz^2 \hat{k}$ at the point (1, 1, 1) (5)

b) Find the curvature and torsion for the curve (5)

$$x = a \sin t, y = a \cos t, z = at \cot \alpha$$

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] MARCH 2023

Paper Code: BS-111

Subject: Applied Mathematics-I

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt **all** questions:-

- (a) If $\int_0^1 x^m dx = \frac{1}{m+1}$, then find the value of $\int_0^1 x^m (\log x) dx$. (2.5)
- (b) If $\vec{F} = (x + y + 1)\hat{i} + \hat{j} - (x + y)\hat{k}$, then compute the value of $\vec{F} \cdot \text{Curl}(\vec{F})$. (2.5)
- (c) Find the particular integral for the linear differential equation: (2.5)

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = \sin 3x.$$

- (d) Determine the rank of the matrix $A = \begin{bmatrix} 1 & 3 & 2 & 1 \\ 1 & 2 & 5 & 2 \\ 2 & 1 & 1 & 3 \end{bmatrix}$. (2.5)

(e) Applying Gauss divergence theorem, find the value of $\int_S \vec{F} \cdot \hat{n} ds$, for $\vec{F} =$

$$(x^2 - yz)\hat{i} + (y^2 - xz)\hat{j} + (z^2 - yx)\hat{k} \text{ and } S \text{ is the cube } 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1.$$

(2.5)

- (f) Find the stationary values of the function $f(x, y) = x^3y^2(1 - x - y)$. (2.5)

UNIT-I

- Q2 (a) If $u = f(r)$ and $x = r \cos(\theta)$, $y = r \sin(\theta)$, then prove that (7)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$$

- (b) Evaluate $\int_0^a \frac{\log(1+ax)}{1+x^2} dx$ and hence show that $\int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi}{8} \log 2$. (8)

- Q3 (a) Find the shortest and longest distances from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$. (6)

- (b) If $u = x^2 - y^2$, $v = 2xy$ and $x = r \cos(\theta)$, $y = r \sin(\theta)$, then find the Jaccobian $J = \frac{\partial(u, v)}{\partial(r, \theta)}$. (4)

- (c) If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. (5)

UNIT-II

- Q4 (a) Solve the ordinary differential equation: $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = \log x \sin(\log x)$. (8)

- (b) Prove that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$, where $J_n(x)$ represents the Bessel function of first kind. (7)

- Q5 (a) Let the electric equipotential lines (curves of constant potentials) between two concentric cylinders be given by $x^2 + y^2 = c$, where c is the constant. Find their orthogonal trajectories (known as curves of electric force). (4)

- (b) Solve the ODE: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \log x$, by variation of parameters. (6)

- (c) Solve the ODE: $2y dx + x(2 \log x - y) dy = 0$, by choosing suitable method. (5)

P.T.O.

UNIT-III

Q6 (a) Test the consistency and solve the system of equations: (7)
 $3x + y + 2z = 3, 2x - 3y - z = -3, x + 2y + z = 4.$

(b) Verify Cayley - Hamilton Theorem for the matrix (8)
 $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$, and hence find A^{-1} .

Q7 (a) Check whether the vectors $(2, 1, 1), (2, 0, -1), (4, 2, 1)$ are Linearly dependent (6)
 or independent?

(b) Reduce the quadratic form $2xy + 2yz + 2zx$ into the canonical form and (9)
 discuss its nature.

UNIT-IV

Q8 (a) What is the directional derivative of $\phi = xy^2 + yz^3$ at the point $(2, -1, 1)$, (7.5)
 in the direction of the normal to the surface $x \log z - y^2 = -4$ at $(-1, 2, 1)$.

(b) Apply Stoke's Theorem to evaluate $\int_C (y dx + z dy + x dz)$, where C is the (7.5)
 curve of intersection of $x^2 + y^2 + z^2 = a^2$ and $x + z = a$.

Q9 (a) Find the curvature and torsion of the Helix $x = a \cos t, y = a \sin t, z = bt$. (7.5)
 (b) Prove that $\text{div}(\text{grad } r^n) = \nabla^2(r^n) = n(n+1)r^{n-2}$, where $r = \sqrt{(x^2 + y^2 + z^2)}$. (7.5)

END TERM EXAMINATION

FIRST SEMESTER (B. TECH) JANURARY-2024

Paper Code: BS-111

Subject: Applied Mathematics-I

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1. a) Find $\frac{dy}{dx}$ if $(\cos x)^y - (\sin y)^x = 0$ (3)

b) Prove that $A^3 - 4A^2 - 3A + 11I = 0$, where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$ (3)

c) Show that the differential $df = x^2 dy - (y^2 + xy) dx$ is not exact, but that $dg = (xy^2)^{-1} df$ is exact (3)

d) Find the directional derivative of $f(x, y, z) = xy^3 + yz^2$ at the point $(2, -1, 1)$ in the direction of vector $\hat{i} + 2\hat{j} + 3\hat{k}$ (3)

UNIT-I

Q2. a) If $u = f(r)$ where $r^2 = x^2 + y^2 + z^2$, show that $u_{xx} + u_{yy} + u_{zz} = f''(r) + \frac{2}{r} f'(r)$ (4)

b) If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$ (4)

c) Locate the stationary points of the function $f(x, y) = (x^2 - 2y^2) \exp\left[-(x^2 + y^2)/a^2\right]$ where a is a non-zero constant (4)

Q3. a) If $u = x^2 - y^2, v = 2xy$ and $x = r \cos \theta, y = r \sin \theta$, find $\frac{\partial(u, v)}{\partial(r, \theta)}$ (4)

b) The temperature of a point (x, y) on a unit circle is given by $T(x, y) = 1 + xy$. Find the temperature of the two hottest points on the circle. (5)

c) Evaluate $\int_0^{\infty} \frac{e^{-ax} \sin bx}{x} dx$ by applying differentiation under integral sign (3)

UNIT-II

Q4. a) Solve $(1 - x^2) \frac{dy}{dx} + xy = xy^2$ (5)

b) Find the orthogonal trajectories of the family of parabola $y = ax^2$ (2)

c) Solve $\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$ (5)

P.T.O.

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- Q5. a) Find the complete solution for the differential equation $(D^2 - 2D + 1)y = xe^x \sin x$ (5)
 b) Show that the solution of $y'' - 2y' + 2y = 0$ are linearly independent (2)
 c) Apply the method of variation of parameters, to solve $x^2 y'' + xy' - y = x^2 e^x$ (5)

UNIT-III

- Q6. a) Determine the rank of the following matrix (4)

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

- b) Discuss the consistency of the following of equations for various values of λ : (5)
 $2x_1 - 3x_2 + 6x_3 - 5x_4 = 3$
 $x_2 - 4x_3 + x_4 = 1$
 $4x_1 - 5x_2 + 8x_3 - 9x_4 = \lambda$
 c) Solve $x + 2y + 3z = 20, 7x + 3y + z = 13, x + 6y + 2z = 0$ by Gauss Elimination Method (3)

- a) State Cayley-Hamilton Theorem and Use it to find the inverse of the matrix (4)
 Q7. $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

- b) Reduce the quadratic form $4xy + 4yz + 4zx$ in to the canonical form and discuss its nature (3)
 c) Solve $x + 2y + 3z = 20, 7x + 3y + z = 13, x + 6y + 2z = 0$ by Cramer's rule (5)

UNIT-IV

- a) Apply Stoke's theorem to evaluate $\int_C (ydx + zdy + xdz)$, where C is the curve of (8)
 Q8. intersection of $x^2 + y^2 + z^2 = b^2$ and $x + z = b$
 b) Show that vector field $\vec{F} = \Delta(x^3 + y^3 + z^3 - 3xyz)$ is irrotational (4)

- a) Find the divergence and curl of $F = xyz \hat{i} + 3x^2 y \hat{j} + (xz^2 - y^2 z) \hat{k}$ at the point (6)
 Q9. $(2, -1, 1)$
 b) Find the curvature and torsion for the curve $x = a \cos t, y = a \sin t, z = at \cot \alpha$ (6)

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

Paper Code: ETMA-101

Subject: Applied Mathematics-I

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

- Q1 Attempt all questions:- (1.2x10=12)
- Expand $\log_e(1+x)$ in power of x by Maclaurin's theorem.
 - Find the percentage error in the area of an ellipse when an error of +1 percent is made in measuring the major and minor axis.
 - Find the asymptotes of the curve $x^2y^2 - x^2y - xy^2 + x + y + 1 = 0$.
 - Find the radius of curvature of the curve $\frac{1}{x^2} + \frac{1}{y^2} = 1$ at $(\frac{1}{4}, \frac{1}{4})$.
 - Define Hermitian matrix with example.
 - Examine the system of vectors are linearly dependent or linearly independent $x_1 = (1, 2, 3)$, $x_2 = (2, -2, 6)$.
 - Solve $(x+1)\frac{dy}{dx} - y = e^x(x+1)^2$.
 - Find $\int_{-1}^1 x^3 p_4(x) dx$.
 - Write Rodrigue's Formula.
 - Find $(2.5, 1.5)$.

UNIT-I

- Q2 (a) If $y = e^{m \cos^{-1} x}$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$ And hence calculate $y_n(0)$. (8)
- (b) Obtain the first four term in the expansion of $\log \sin x$ in power of $(x-3)$ By Taylor theorem. (4)
- Q3 (a) Find whether the series $\frac{x}{1.2} + \frac{x^2}{3.4} + \frac{x^3}{5.6} + \frac{x^4}{7.8} + \dots$ $x > 0$. Is convergent or divergent. (6)
- (b) Prove that the series $\frac{2}{1} + \frac{3}{4} + \frac{4}{9} + \frac{5}{16} + \dots$ is divergent. (6)

UNIT-II

- Q4 (a) Evaluate $\int_0^\infty \sqrt{x} e^{-x^{\frac{1}{3}}} dx$. (6)
- (b) Find the area bounded by the parabola $y^2 = 4ax$ and its latus rectum. (6)
- Q5 (a) Trace the curve $r = a \sin 2\theta$. (8)
- (b) Evaluate $\int_0^\pi \cos^9 \theta d\theta$. (4)

UNIT-III

- Q6 (a) Find the inverse of matrix A by Gauss Jordan method if (6)

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

P.T.O.

- (b) Find the eigen values and eigen vector of matrix (6)

$$A = \begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$$

- Q7 (a) Verify Cayley Hamilton theorem of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence

find The value $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + 1$. (8)

- (b) For which value of "a" the rank of the matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ a & 13 & 10 \end{bmatrix}$ is 2 (4)

UNIT-IV

- Q8 (a) Prove that $J_{\frac{5}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left[\left(\frac{3-x^2}{x^2} \right) \sin x - \frac{3 \cos x}{x} \right]$. (6)

- (b) Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$. (6)

- Q9 (a) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \cos x$. (6)

- (b) Solve $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + 4y = \cos(\log x) + x \sin(\log x)$. (6)

(Please write your Enrolment No. immediately)

Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

1st Semester, October, 2024

Paper Code: BS-111

Time: 1½ Hrs.

Subject: Applied Math-I

Max. Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks	CO(s)
1(a)	If $z = f(x, y)$ and $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, then show that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}.$	2.5	CO1
1(b)	If $u = x^2$, $v = y^2$, find $\frac{\partial(x, y)}{\partial(u, v)}$	2.5	CO1
1(c)	If $z = e^{xy}$, $x = t \cos t$, $y = t \sin t$, find $\frac{dz}{dt}$ at $t = \frac{\pi}{2}$.	2.5	CO1
1(d)	Find the complementary function (C.F.) of $(D^4 - 2^4)(D + 2)y = 0,$	2.5	CO2
2(a)	If $u = f[e^{y-z}, e^{z-x}, e^{x-y}]$, then show that $u_x + u_y + u_z = 0$.	5.0	CO1
2(b)	If $\tan^{-1}\left(\frac{x^3 + y^3}{x - y}\right)$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.	5.0	CO1
3(a)	If $x^2 + y^2 + u^2 - v^2 = 0$ & $uv + xy = 0$, then prove that $\frac{\partial(u, v)}{\partial(x, y)} = \frac{x^2 - y^2}{u^2 + v^2}$	5.0	CO1
3(b)	Find all the stationary points of the functions and examine whether the function is maximum or minimum at those points. $F(x, y) = x^3 + y^3 - 3axy$ ($a > 0$)	5.0	CO1
4(a)	Solve $(y^3 - 3xy^2)dx + (2x^2y - xy^2)dy = 0$	5.0	CO2
4(b)	Find the orthogonal trajectories of the family of curves $x^2 - y^2 = c$	5.0	CO2